



Chapter 2

Mapping QTLs and Identification of Genes Associated with Drought Resistance in Sorghum

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Abstract

Water limits global agricultural production. Increases in global aridity, a growing human population, and the depletion of aquifers will only increase the scarcity of water for agriculture. Water is essential for plant growth and in areas that are prone to drought, the use of drought-resistant crops is a long-term solution for growing more food for more people with less water. Sorghum is well adapted to hot and dry environments and has been used as a dietary staple for millions of people. Increasing the drought resistance in sorghum hybrids with no impact on yield is a continual objective for sorghum breeders. In this review, we describe the loci, quantitative trait loci (QTLs), or genes that have been identified for traits involved in drought avoidance (water-use efficiency, cuticular wax synthesis, trichome development and morphology, root system architecture) and drought tolerance (compatible solutes, pre- and post-flowering drought tolerance). Many of these identified genes and QTL regions have not been tested in hybrids and the effect of these genes, or their interactions, on yield must be understood in normal and drought-stressed conditions to understand the strength and weaknesses of their utility.

Key words Drought avoidance, Drought tolerance, Drought escape, Trichomes, Epicuticular wax, Stay-green, Root architecture, Water-use efficiency, Compatible solutes

1 Introduction of Drought Resistance

Drought, described as soil and/or atmospheric water deficit, has been a major cause of crop loss [1]. With increasing aridity in many areas of the globe and a growing human population, water will likely be a scarcer commodity [2]. Thus the utilization of crop cultivars adapted to water-stressed conditions is a long-term solution for stabilizing crop productivity [3]. Water is essential for plant growth and serves many roles such as a solvent, a reactant, and a maintainer of cellular turgidity [4]. Thus, water limitation causes plants to halt cell enlargement, leaf and stem elongation, and photosynthesis and to decrease respiration and turgor, which increases wilting [4]. Drought can occur at any plant growth stage and can affect crop productivity depending on the time of

onset, duration, and intensity of drought stress [5]. Drought stress during the seedling or reproductive stages can result in stand losses or severe yield losses [6, 7].

Drought resistance is defined as, under the same drought intensity, certain genotypes of crop plants having higher relative yield as compared with others [8]. According to Levitt [9], plants that are resistant to drought physiologically use three strategies that include escape, avoidance, and tolerance strategies of which plants may combine these responses [10]. Plants that escape drought can complete their life cycle before physiological water deficits occur and exhibit developmental plasticity [2]. Desert plants exhibit high stomatal conductance for carbon dioxide and photosynthetic efficiency, rapid growth and development, and can complete their life cycles in a few weeks [11]. Additionally, in another example of drought escape, the use of early maturing genotypes or the timing of the planting of crops can be chosen, so that during the growing season, local seasonal or climatic droughts are avoided [10, 12].

Drought avoidance refers to plants that can endure drought by avoiding tissue dehydration [2]. This involves minimizing water loss from transpiration or the cuticle and/or maximizing water uptake by the roots [2]. Water loss is minimized by closing stomata [4], by the excessive deposition of epicuticular wax on the aerial surface of the plant that prevents cuticular transpiration [13], and by the presence of trichomes that entrap and retain surface water allowing foliar water uptake [14]. Other drought avoidance mechanisms include leaf abscission [15], dormancy [16], and leaf rolling to reduce water loss from transpiration [17]. Additionally, plants that use C_4 photosynthesis have less water loss from transpiration than C_3 plants [18]. Furthermore, plants possessing deep, large root systems that can penetrate hard soil layers are often associated with the ability to maintain water supply during low water conditions as well as the ability of the root system to respond to water stress [19]. Under water stress, increased root/shoot ratios are often seen, and the root/shoot ratios have been used as a criterion to identify drought resistance in plants [10]. The ability of some plants to store water using specific organs (tubers, etc.) is also a drought avoidance trait [20].

Drought tolerance mechanisms allow the plant to maintain metabolic activity under low tissue water potential by osmotic adjustment and increased antioxidant defense [3, 21, 22]. Plants can accumulate compatible solutes such as sugars and sugar alcohols (fructan, trehalose, mannitol, d-ononitol, and sorbitol), ammonium compounds (polyamines, glycinebetaine, *b*-alanine betaine, dimethyl-sulfonio propionate, and choline-*O*-sulfate), and amino acids (proline and ectoine) which accumulate in the cytosol in order to lower osmotic potential and maintain the turgor of both shoots and roots [23]. During drought, reactive oxygen species (ROS) generation is increased due to the disruption of the

electron transport system [24]. ROS are damaging to cellular components and can cause lipid peroxidation or protein and nucleic acid modification [24]. Antioxidants allow plants to detoxify ROS by using antioxidant enzymes and free radical scavengers [25]. Antioxidant enzymes include superoxide dismutase (SOD), catalases, peroxidases, glutathione S-transferase, glutathione reductase, monodehydro/dehydroascorbate reductase, alternative oxidase, etc. and the free radical scavengers are carotenoids, ascorbate, tocopherols, cytochrome f, flavanones, anthocyanins, glutathione, etc. [10, 25]. Thus, osmotic adjustment and the alleviation of drought damage are mechanisms of drought tolerance.

2 Crosstalk Between Abiotic and Biotic Stress Responses

In the past, the molecular mechanisms of each abiotic and biotic stress were examined independently but plants respond to abiotic and biotic stresses with common and overlapping pathways [26] which will be briefly reviewed here. Abiotic and biotic stresses activate the hormone signaling pathways regulated by abscisic acid (ABA), salicylic acid, jasmonic acid, and ethylene, generate ROS, and cause mitogen-activated protein kinase (MAPK) cascades [26]. To reduce the effects of the stress, plants use hormone signaling, ROS accumulation, and MAPK cascades to activate transcription factors that cause the expression of downstream response genes [27].

Plant hormone signaling controls abiotic and biotic stress responses [26]. Abiotic stress is largely controlled by ABA whereas biotic stress is controlled by the antagonism between salicylic acid and jasmonic acid/ethylene signaling pathways [27]. Additionally, ABA can increase the susceptibility of plants to pathogens and ABA insensitive lines have higher levels of pathogen resistance than wild type plants [28]. ABA can also have a positive effect on pathogen defense but the timing of infection and the type of pathogen affects the influence of ABA [29].

ROS accumulate constantly as by-products of metabolism but their concentrations increase to high levels after abiotic stress [27]. Additionally, plants under pathogen infection induce a hypersensitive reaction and/or systemic acquired resistance activation which leads to ROS generation [30]. The generation of ROS after abiotic and biotic stress then causes plants, in order to prevent damage to membrane phospholipids and fatty acids, to induce large sets of genes that encode ROS-scavenging enzymes or antioxidants [31].

MAPK cascades transduce the perception of the environmental and developmental cues into internal signaling pathways [32]. During biotic stress, plant transmembrane receptors detect pathogen-associated molecular patterns and trigger MAPK cascades to start

pathogen response signaling [33]. Likewise, during abiotic stress, MAPK cascades are induced following activation of receptors [32]. Downstream targets of MAPK cascades include transcription factors, other kinases, enzymes, and hormone response factors [32, 34].

Transcription factors control a wide range of downstream events and many transcription factors have been found to be involved in the response to both biotic and abiotic stress [27]. The transcription factors Myelocytomatosis Oncogene2/Jasmonate Insensitive1 (*MYC2/JIN1*), the myeloblastosis (MYB) family of transcription factors; No Apical Meristem (NAM), Arabidopsis Transcription Activation Factor (ATAF), and Cup-shaped Cotyledon (CUC) of which these three are members of one of the largest families of plant-specific transcription factors and are abbreviated (NAC); APETALA2/Ethylene Responsive Factor (AP2/ERF), ABA-responsive Element-binding Factor (ABF), transcription factors defined by the conserved WRKYGQK sequence (WRKY), Zinc-fingers, and heat shock transcription factors all have been shown to have a role in the interaction between abiotic and biotic stress signaling pathways [27]. Heat shock transcription factors control many downstream stress-responsive genes as well as the expression of heat shock proteins which bind and stabilize proteins that have become denatured after abiotic and biotic stress [35]. Each stress may elicit activation of a unique combination of heat shock proteins [27]. Heat shock transcription factors may detect the presence of ROS and cause the expression of heat shock proteins as well as ROS scavenging enzymes [35].

Compilation of drought responses, specifically, in a range of plant species reveals that drought induces hormone signaling by ABA, a hormone involved in stomata closure to reduce transpiration [36]. Drought suppresses cell growth and photosynthesis efficiency, decreases respiration [4], and induces genes that are classified into two groups [36]. The first group includes proteins such as chaperones, late embryogenesis abundant (LEA) proteins, osmotin, mRNA-binding proteins, osmolyte biosynthesis, water channel proteins, metabolite transporters, detoxification enzymes, and proteases which all function in abiotic stress tolerance [36]. The second group involves proteins involved in signal transduction and stress-responsive gene expression and includes kinases, protein phosphatases, calmodulin-binding proteins, and enzymes involved in phospholipid metabolism [36].

3 Sorghum

Sorghum (*Sorghum bicolor* L. Moench) is a C₄ annual grass originating from Africa [37] that is used as a grain crop, for forage, and as a sweetener [38]. It also has potential to be used as a bioenergy

feedstock [39]. Sorghum has drought and heat tolerance and is grown in Africa, Asia, the Americas, and Australia in regions that are typically too dry for maize (*Zea mays* L.) production [40, 41]. The drought resistance seen in sorghum is attributed to morphological and physiological factors such as a root system whose depth is dependent on the amount of water received [42], the ability to maintain stomata open and thus continue photosynthesis at low water potentials, leaf senescence, and the ability for osmotic adjustment [43]. Furthermore, high epicuticular wax deposition on the sorghum plant surface reflects visible and near-infrared radiation and reduces cuticular conductance of water vapor [44]. In comparison to maize, sorghum has a higher water-use efficiency [45]. Sorghum is consumed as a dietary staple in parts of Africa and Asia and is used predominantly as a poultry/animal feed, with only a small amount being used for human consumption in the United States and Australia [46]. Consumption of sorghum provides energy, vitamins, and minerals, and the grain has a protein content of near 9% [47].

In the United States from 2000 to 2016, the average acres of planted grain sorghum were 7.65×10^6 acres [47]. In the U.S., sorghum is grown primarily from South Dakota to Southern Texas (the Sorghum Belt) [48]. Most of the sorghum grown in the U.S., for the last few years, has been exported; during 2013–2014, China imported four million metric tons of sorghum from the U.S [49]. China has been importing sorghum as an alternative to corn as there is no import quota on sorghum [50]. This demand from China has a strong, positive impact on the world and U.S. grain sorghum prices and farm production [47].

4 Molecular Mechanisms of Drought Responses in Sorghum

With the crosstalk of the abiotic and biotic signaling pathways as a framework, we will now briefly cover the literature on the molecular mechanisms of drought responses in sorghum. Transcript-profiling studies are beginning to elucidate the roles that various genes play in adaptation to drought. As part of a larger sorghum transcriptome project, three drought-induced expressed sequence tag (EST) libraries were constructed [51]. For the first library-water-stress 1 (WS1), five-week-old whole greenhouse-grown plants (including roots) of inbred line BTx623 were harvested after water was withheld for 7–8 days. Two libraries were created from the leaves of greenhouse-grown plants at either pre-flowering drought stress (DSBF1) or post-flowering drought stress (DSAF1). The pre-flowering drought-tolerant line Tx7000 was used to create DSBF1, while the stay-green post-flowering drought-tolerant line B35 (a.k.a. BTx642) was used to create the DSAF1 library. The DSBF1 and DSAF1 libraries were subtracted using cDNA from

unstressed sorghum leaves to enrich these libraries with transcripts preferentially expressed under drought stress [52]. Dehydrins, heat-shock proteins, and other drought-inducible proteins of unknown function are among the transcripts expressed in these libraries. Fifty-five of these transcripts were also found to be upregulated by ABA in a microarray experiment highlighting the role of ABA signaling in the drought response [53]. The software program Simple Sequence Repeat Identification Tools (SSRIT) [54] was used to search for simple sequence repeat (SSR) sequences within the ESTs from the DSAF1 and DSBF1 libraries [55]. From over 9800 ESTs, 109 SSR markers were developed. Using 168 recombinant inbred lines (RILs) from a cross between 298B and IS18551, 28 of these markers were placed on a linkage map, along with 100 genomic SSRs. Following the publication of the *Sorghum bicolor* genome sequence [56] and the development of high-throughput sequencing methods, RNA-Seq was used to re-examine the transcriptomes of sorghum plants exposed to osmotic stress induced by polyethylene glycol (PEG) or ABA [57]. They found that about 1650 genes were upregulated by osmotic stress, and about 700 genes were downregulated. Exogenous ABA caused upregulation of about 2300 genes and downregulation of about 2600 genes. Depending on the tissue examined (leaf or root), between 12 and 30% of these genes showed similar regulation by osmotic stress and ABA. Genes that were upregulated by osmotic stress and ABA included a dehydrin and a late-embryogenesis-abundant (LEA) protein. Downregulated genes included a gene with sugar transporter domains in the root tissue and a peroxidase-like gene in the shoot tissue. Pathway analysis revealed that genes involved in synthesis of sugars, amino acids, and other osmoprotectant molecules were also upregulated by osmotic stress and ABA. Several pathways involved in hormone biosynthesis or regulation were also affected.

A transcriptome study using microarrays was conducted on sorghum seedlings subjected to heat, drought, or heat and drought in combination, as for summer-grown crops the two stresses are often simultaneous [58]. For drought stress only, genes involved in water deprivation, response to ABA, regulation of photosynthesis, wax biosynthesis, fluid transport, and amino acid metabolism were induced. Examination of the promoter motifs for the 966 induced genes revealed the most highly represented promoter motif was similar to the abscisic acid response element (ABRE): (C/T) ACGTGTC. Examination of the genes induced by drought with a previous study that used PEG [57] revealed that 32% of the induced genes were shared between the two studies. For heat and drought stress, genes regulated by the combined stresses encode ion transporters, heat shock proteins, LEA proteins, signaling proteins, calcium binding proteins, MAP kinases, and transcription factors. Ontology analysis revealed an enrichment of genes involved in

polyamine metabolism and, in particular, spermidine biosynthesis which may be involved in reactive oxygen scavenging and membrane protection [59]. The most enriched promoter motif was similar to the ABRE, which was also the promoter motif for drought stress only. This study reveals although many common pathways and genes are regulated by heat stress, PEG, drought stress, and a combination of heat and drought stress, there are genes, and thus pathways, that are uniquely altered by each particular stress.

The sorghum proteome responses to drought stress and recovery from drought have also been examined at the fifth leaf stage using matrix-assisted laser desorption/ionization-time-of-flight mass spectrometry (MALDI-TOF-MS) [36]; for the principles and technical aspects of MALDI-TOF-MS see Kasper et al. [60]. Proteins that were differentially regulated after drought and/or recovery were homologs potentially involved in (1) metabolism esp. methionine synthase, (2) energy esp. C₄ phosphoenolpyruvate carboxylase and chloroplast NADP-dependent malic enzyme, fructose-1,6-bisphosphate aldolase, and glyceraldehyde-3-phosphate dehydrogenase, (3) transcription esp. RNA-binding proteins, (4) protein synthesis of which one protein was differentially regulated, elongation factor alpha, and (5) chaperones esp. heat shock protein 60. The authors conclude that the drought-tolerant genotype of sorghum maintained protein stability, reallocated metabolites to the newly developed structures, and maintained protein synthesis while the drought susceptible genotype had more proteins involved in cell death and proteases [36].

5 QTL Mapping

Many agriculturally important traits such as yield and drought response are controlled by many genes and are referred to as quantitative traits [61]. QTL are sections of DNA, usually DNA markers, which correlate with variation in a phenotype. QTL are typically identified by bi-parental mapping where a linkage map, made up of ordered DNA markers, is created and the parents (which differ in the traits of interest) and progeny are phenotyped in multiple locations and/or years. QTL are identified using single marker analysis, simple interval mapping, and composite interval mapping, of which, composite interval mapping is currently the most frequently used, reviewed in detail by Collard et al. in 2005 [61]. An alternative approach to QTL mapping is association mapping, which uses largely unrelated individuals as the population and this population is phenotyped in multiple environments and years and is genotyped [62]. Correlations are identified between genetic markers and phenotypes. Association mapping has higher

mapping resolution than linkage mapping due to the larger number of recombination events in a sample [62].

Many QTL studies have been performed in sorghum due to the diploid tractable nature of the genome, the ease of population development and obtaining germplasm, and the advent of high-throughput sequencing and genotyping-by-sequencing [63]. Furthermore, a reference genome sequence of sorghum [56] and a collection of mutants are also available [64].

6 Sorghum QTLs and Identification of Genes

Drought resistance in sorghum, like many plants, is controlled by numerous small effect loci and hundreds of genes that affect morphological and physiological responses to drought [5]. In this section, we will review the morphological and physiological traits associated with drought resistance that have been examined in sorghum and the QTL studies performed on sorghum to date that are involved in drought resistance as well as the candidate genes identified. The traits measured to assess drought resistance can be classified as drought avoidance-related, drought tolerance-related, and integrative indicators [10].

6.1 Drought Avoidance Traits

6.1.1 Transpiration and Water Use Efficiency

Water use efficiency (WUE) can be defined as the amount of biomass produced divided by the amount of water used, or transpired, by the plant [65]. An indirect way to measure WUE is to measure the CO₂ assimilation rate (A) and the transpiration rate (E), both of which can be measured simultaneously using a portable infrared gas analyzer, such as the Li-Cor LI-6400 Portable Photosynthesis System (Li-Cor, Lincoln, NE), or similar apparatus [66, 67]. The transpiration ratio (A:E) is then calculated as (A/E)/VPD, where VPD is the vapor pressure deficit of water [67]. Water use efficiency is often equated with drought tolerance, but Blum [65] cautions that this is often not true, and that selection for WUE may result in a shorter growing cycle and lower yields.

In two greenhouse studies, A, E, and A:E were measured in a population of 70 RILs (Tx430 × Tx7078) [67]. The soil moisture in the pots was maintained at either 40% or 80% field capacity (FC) in Experiment 1, and at 80% FC only in Experiment 2. A genetic map was constructed from 261 digital genotyping markers [68]. Composite interval mapping identified one QTL (SBI-05) for A at 40% FC, and one (SBI-01) at 80% FC in the first experiment, and a different QTL (SBI-09) for A at 80% FC in the second experiment (Table 1). Three QTLs were identified for E, two on SBI-01 at different locations, and another on SBI-07. Three QTLs for transpiration ratio at 80% FC were found on SBI-09 (2) and SBI-10 in Experiment 1, and another on SBI-06 at 80% FC in Experiment 2. No QTL for A:E were found for the 40% FC

Table 1
QTLs identified in *Sorghum bicolor* that are associated with drought resistance

Trait	Mapping population	No. of progeny	Chr./ LG ^a	Flanking markers	PV ^b	References
<i>Drought avoidance</i>						
Transpiration and water use efficiency						
CO ₂ assimilation rate (A)	Tx430 × Tx7078 F ₆	70	1	DGA18–DGA20	12	[67]
			5	DGA216–DGA219	15	
			9	DGA40–DGA41	11	
Transpiration rate (E)	Tx430 × Tx7078 F ₆	70	1	DGA29–DGA30	12	[67]
			1	DGA20	15	
			7	DGA291–DGA293	13	
A:E ratio	Tx430 × Tx7078 F ₆	70	6	DGA6–DGA47	12	[67]
			9	DGA326–DGA327	19	
			9	DGA329–DGA330	21	
			10	DGA395–DGA398	17	
Stomatal conductance	Early Hegari-Sart × BK7 F _{3:4} , 4:5	28,107	7	58,192–58,594 kb	4	[69]
			10	0–781 kb	1	
Stomatal density	Tx430 × Tx7078 F ₆	70	2	DGA89	17	
			7	DGA293–DGA294	15	
<i>The cuticle</i>						
Cutin/epicuticular wax	blmc × BTx623 F ₂	220	10	Xsbarsl10.47, Xcup42	– ^c	[89]
Epicuticular wax	BTx623 × RS647 F ₂ /F ₇	605/116	1	Sblm13/Sblm16	–	[81]
<i>Trichomes</i>						
Trichome density-upper	296B × IS18551 F ₇	168	10	Xnhsbm1044-Xnhsbm1013	16	[102]
			10	Xnhsbm1043-Xgap1	33	
Trichome density-lower	296B × IS18551 F ₇	168	1	Xtxp32-Xtxp88	9	[102]
			1	Dsenhsbm80-xtxp302	5	
			4	Xcup48-Ungnhsbm32	9	
			6	Xtxp317-Xtxp274	5	
			10	Xnhsbm1048-Xnhsbm1013	15	
			10	Xnhsbm1043-Xgap1	23	

(continued)

Table 1
(continued)

Trait	Mapping population	No. of progeny	Chr./ LG ^a	Flanking markers	PV ^b	References
<i>Root systems</i>						
Crown root angle, mature	Early Hegari-Sart × BK7 F _{3:4} , 4:5	28,107	3	4631–5166 kb	3	[69]
Nodal root angle, seedling	BR23296 × SC170-6-8 F ₆	141	5	SPb-5892–SPb-4323	10	[108]
			5	SPb-6287–SPb-9490	30	
			8	SPb-4767–SPb-4432	7	
			10	SPb-1660–SPb-7058	12	
Nodes with brace roots	Sansui × Jiliang F ₂	611	6	Xtxp127–Xtxp6	7	[111]
			7	Dsenhsbm7–Xcup70	53	
Root dry weight, seedling	BR23296 × SC170-6-8 F ₆	141	2	SPb-4366–SPb-5544	13	[108]
			5	SPb-4086–SPb-6323	12	
			8	SPb-7889–SPb-6935	7	
Root length, mature	E36-1 × SPV570 F ₈	184	4	Xsnp56–Xiabt194	8	[110]
Roots/plant, mature	E36-1 × SPV570 F ₈	184	1	Xiabt210–Xiabt69	18	[110]
Root:shoot ratio, mature	E36-1 × SPV570 F ₈	184	10	Xiabt489–Xiabt364	8	[110]
			10	Xiabt312–Xiabt178	8	
Root volume, mature	E36-1 × SPV570 F ₈	184	1	Xiabt210–Xiabt69	14	[110]
			4	Xtxp51–Xtxp270	13	
Root weight, mature	E36-1 × SPV570 F ₈	184	4	Xtxp51–Xtxp270	9	[110]
<i>Drought tolerance</i>						
<i>Pre-flowering</i>						
Pre-flowering DT ^d (height stability)	Tx7078 × B35 F ₅	98	A	bB13/35	43 ^c	[125]
Pre-flowering DT (Prf F)	SC56 × Tx7000 F ₇	125	F	psb553-psb708	22.2	[127]

(continued)

Table 1
(continued)

Trait	Mapping population	No. of progeny	Chr./ LG ^a	Flanking markers	PV ^b	References
Pre-flowering DT (Prf E)			E	txs1075a-psb758	11.9	
Pre-flowering DT (Prf G)			G	txs645-hhu504	15	
Pre-flowering DT (seed stability)			E	b258/94	14.2 ^c	
Pre-flowering DT (yield per se)			H	b374/75-t380/67	42.1 ^c	
			D	UMC85	42.1 ^c	
Pre-flowering DT (Yield stability)			F	tC13/150	38.7 ^c	
<i>Post-flowering (stay-green)</i>						
Stay-green (<i>stg1</i>)	BTx642 × Tx7000 F ₇	97	3	NPI1414-BNL15.20	17.1	[41, 147]
Stay-green (<i>stg2</i>)			3	Xtxs584-A12-420	24.5	
Stay-green (<i>stg3</i>)			2	TXS1307-UMC5	10.7	
Stay-green (<i>stg4</i>)			5	TXS713-rcb	11.2	
Stay-green (<i>stg5</i>)			1	CPY79A1-UGT85B1	10.8	[149]
% dry leaves after drought	Association panel	107	1	Xtxp149	19–20	[151]

^aLG-linkage group^bPV-phenotypic variation^cSingle nuclear recessive gene^dDT-drought tolerance^ePV was combined

treatment. Stomatal density was measured in Experiment 2, and a QTL for this trait was found on SBI-02. The correlation coefficient (R^2) values for these QTLs ranged between 11 and 21%. This same population was also planted in a field experiment, and stomatal density was again measured. This time a QTL was found on SBI-07, with an R^2 of 15%.

Two QTLs for stomatal conductance on SBI-07 and SBI-10 in an Early Hegari-Sart × BK7 population were identified [69] (Table 1). During the growing season, stomatal conductance (g_s) was measured using a Li-Cor LI-6400XT. Further investigation of the population contrasting for the QTL on SBI-07 showed that the difference in stomatal conductance was not due to stomata number

or pore size. There was also no negative trade-off between reduced stomatal conductance and CO₂ assimilation rate (A). The authors hypothesized that this QTL may function in regulation of stomatal closure.

Aquaporins are membrane-bound proteins which allow water, and sometimes other small molecules, to pass through cell and organelle membranes [70]. In drought-stressed rice (*Oryza sativa* L.) seedlings, it was found that aquaporins are the main route through which water enters the plant, as opposed to the apoplastic route [71], and thus their role in conferring drought tolerance in sorghum merits investigation. The roles of four aquaporin genes in field-grown maize and sorghum under drought versus well-watered conditions were studied [72]. As expected, the drought treatment reduced shoot growth in both species, but had a much greater effect on maize. Relative water content of maize was also reduced by the drought treatment, but it was unaffected in sorghum. The aquaporin gene *PIP2;3* was more highly expressed in root tissue under drought stress than under control (well-watered) conditions in the predawn and early morning hours in sorghum. A similar pattern was observed for aquaporin gene *TIP1;2* in sorghum roots, except that expression in the control increased in the late afternoon. These results suggest that these genes play an important role in water uptake by sorghum roots under drought, though more studies are needed. Drought also appeared to induce expression of aquaporin gene *PIP1;5* in drought-stressed sorghum leaves; expression was also detected in roots but the pattern was inconsistent. In hydroponically grown sorghum seedlings, addition of PEG to the culture medium induced osmotic stress, and caused upregulation of several aquaporin genes in the roots [73]. Treatment with silicon (Na₂SiO₃) also caused upregulation of several aquaporins, suggesting a possible role of this element in signaling of osmotic stress.

6.1.2 The Cuticle

The plant cuticle covers the above ground organs of all land plants providing a hydrophobic surface layer or “protective skin” [74, 75]. The cuticle is composed of a framework of cutin and integrated and superimposed lipids; the integrated waxes in the cuticle are called intracuticular waxes while the superimposed lipids are called epicuticular waxes [74]. The cuticle serves primarily as a barrier to transpirational water loss, allowing gas exchange and transpiration to be controlled by the stomata (a mechanism of drought avoidance) but also has a role in signaling, plant defense, and development [74, 75]. Cuticles are not impermeable and depending on the plant species, have permeances for water [76], chemicals [77], and gases [78, 79].

Epicuticular waxes (EWs) coat the aerial surface of sorghum plants and appear as a white soft material which is prominent at the boot leaf stage on the leaf sheath and the lower part of the leaf blade

[80, 81]. EWs reduce cuticular conductance and reflect incoming radiation at the UV and 400–700 nm wavelengths to reduce leaf temperature and transpiration [65]. Epi- and intra-cuticular waxes from sorghum sheaths are made up of primarily fatty acids (96%), of which the C28 and C30 acids composed 77% and 20% of these fatty acids, respectively [82]. The very long fatty acid acyl chains that are wax precursors are activated by coenzyme A (CoA) and synthesized by elongases; these elongases use malonyl-CoA as the two-carbon donor [83]. Two epidermal cell types produce epi-cuticular wax crystals on sorghum leaf sheaths; epidermal cork cells and epidermal long cells produce tubular filament wax crystals and plate-like waxes, respectively [84]. In sorghum, epicuticular wax mutants are designated as bloomless (lacking visible epicuticular wax) or sparse bloom (reduced visible wax) [85]. Jordan et al. [86] estimated that in sorghum EW load of about 0.067 g m^{-2} could be effective in preventing water loss through the cuticle. To determine the precise contribution of single EW loci, diethyl sulfate (DES) and ethyl methanesulfonate (EMS) were used on two drought-resistant sorghum inbreds (P954035 and P898012) to generate 33 M_2 generation mutants with altered EW [85]. Scanning electron microscopy was used to group the lines into 14 classes based on altered EW structure [85]. Furthermore, 12 of these bloomless (*bm*) and sparse bloom (*h*) mutants were examined for the amount and composition of cuticular waxes on the sheaths of the mutants and were found to have significant reductions in the amount of C28 and C30 fatty acids as compared to wild type plants [82]. Using this collection of mutants and some mutants previously described [87], an allelism study was conducted, and 10 *bm* and 10 *h* loci were identified [80]. Sorghum lines with mutations in the *Bm2* locus lack EW and have thinner and lighter cuticles [84]. Furthermore, these *bm2* mutants have rapid leaf water loss rates, leakier epidermal layers, and higher susceptibility to a fungal pathogen [84, 88]. Genetic mapping of the *bm2* locus using the mutated line *blmc* [89] (Table 1), which has been renamed as *bm2-7* [90], has been found to be on sorghum chromosome 10. Identification of the *Bm2* gene will help elucidate the cuticle/EW pathways in sorghum and its role in improved abiotic and biotic stress tolerance.

Recently, a sorghum EW locus derived from a spontaneous mutant RS647 was identified [81]. Genetic mapping using BTx623 $\times$ RS647 F_2 and F_7 populations identified a locus *bm39* to a region on SBI-01 (Table 1). A gene glycine-aspartic (glutamic) serine leucine (GDSL)-like lipase/acylhydrolase was suggested as the candidate gene and mRNA levels were lower in the transcript of bloomless near isogenic lines (NILs) as compared to those having bloom [81]. Furthermore, a subgroup of GDSL lipases which has a motif consensus amino acid sequence of Ser, Gly, Asn, and His

(SGNH) hydrolase is associated with hydrolase ability in the leaf epidermis in *Agave americana* L. [91].

Additionally, in an unrelated study, a bloomless sorghum mutant, 2I-400-2-11-3, was generated by gamma irradiation [92]. RNA sequencing was used to identify the genes that were differentially expressed between the wild type and mutant plants in the leaf sheaths, and the most differentially expressed gene, *Sb06g023280*, was similar to an ABC transporter responsible for wax secretion in *Arabidopsis*. Additionally, a 1.4 Mb genomic inversion was identified proximal to the *Sb06g023280* promoter region in the bloomless mutant [92]. Thus, the mutation may have prevented the secretion of wax and this study identifies a gene likely involved in wax secretion in sorghum.

6.1.3 Trichomes

Plants with increased drought resistance often have more epidermal trichomes [10]. Trichomes are specialized epidermal cells that include hairs and scales which protrude above the epidermis of which the hairs may be glandular and able to produce a secretion [93, 94]. Trichomes function in plants to control water and to prevent insect and herbivore damage [95–97]. In many plant species, dense hairs reflect sunlight and reduce the heat load over the leaf surface protecting the tissue in hot and dry environments [98]. Thus, less light absorption reduces leaf temperature and thus reduces the leaf transpiration rate [95]. Additionally, dense trichome canopies can hold droplets of water above the leaf surface to prevent the interference with stomatal pores [99]. Trichome density can be increased in plants after mechanical damage or water shortage [95, 100]. Having trichomes may come at a cost to plants. At low wind speeds, trichomes may increase the turbulence in the boundary layer and increase photosynthetic gas exchange, leading to increased transpiration [101].

Trichomes also serve as a mechanical barrier to insect movement and feeding [102]. Chemical secretions of terpenes, phenolics, alkaloids, or other substances from glandular trichomes can deter or be toxic to herbivores or insects [96, 97]. Hooked trichomes may impale adults or larvae of insects and stinging trichomes can deter large animals [96].

For most studies using sorghum, trichomes have been examined not for their control of water but rather in relation to their effects on pest feeding. Satish et al. [102] performed a QTL study examining trichomes and sorghum shoot fly resistance (*Atherigona soccata* (Rondani) using a 296B (susceptible) × IS18551 (resistant) RIL population. The authors noted that in sorghum the shoot fly-resistant lines often had pointed unicellular trichomes as compared to blunt bicellular trichomes. They mapped this morphological trait as the resistant parent, IS18551, had unicellular pointed trichomes and the susceptible parent had bicellular blunt

trichomes. The gene segregating for trichome morphology mapped as a morphological marker on SBI-10, of which QTLs for shoot fly egg number, shoot fly damage, and trichome density on the upper and lower leaf co-located with trichome morphology. Additional QTLs for trichome density on the upper and lower surfaces of the leaf in the same mapping population were identified on SBI-01, SBI-04, SBI-06, and SBI-10 [102] (Table 1). Two candidate genes for trichome morphology were a fibroin heavy chain and a tubulin beta-2/beta-3 chain which have been found in other species to be specifically expressed in leaf trichomes of tomato (*Solanum lycopersicum* L.) and *Arabidopsis* [102].

6.1.4 Root Systems

Many efforts to phenotype drought tolerance traits in plants have focused on the above-ground portion of the plant, because it is much easier to observe than the roots. However, as the major point of entry for water into the plant, the importance of roots should not be overlooked. Because roots are so difficult to phenotype in the field, root-related traits are ideal candidates for improvement through marker-assisted selection. To date, relatively few studies have identified QTLs associated with root traits in sorghum, but methods to more accurately and quickly phenotype roots are being developed, which should lead to better selection for root-related traits.

In sorghum, root system development begins as the radicle emerges from the seed and elongates to become the primary root [103]. Around the fourth-leaf stage, nodal roots begin to emerge from the first node of the coleoptile, and eventually these will form the bulk of the root system [103]. These nodal roots are called crown roots if they form below ground and are called brace roots if they arise from above-ground nodes [104]. The angle at which the nodal roots grow has been proposed as a trait on which to select for drought avoidance [69, 105]. In theory, a narrower (more vertical) root angle should indicate deeper root growth, and thus better ability to access deeper soil layers where more water is available [69]. In sorghum, the plant needs to grow to about the sixth-leaf stage before the root system is developed enough to observe differences in nodal root angle [103]. This contrasts with other grasses like maize which develop nodal roots much sooner [103]. Hydroponic [106] or germination paper-based methods [107] have been developed for observation of maize roots, but they may not be ideal for sorghum, because these media cannot support the plant long enough [103]. To observe sorghum root angle in-situ, a soil-based system is preferable. Custom root growth chambers were constructed, essentially miniature rhizotrons [108]. They consist of two sheets of clear acrylic 45 × 50 cm, separated on three sides by 3 mm-thick rubber strips, and held together with clamps. The space between the sheets is filled with soil. Seeds can be planted

directly into the chambers, or pre-germinated and then transplanted. The chambers are placed vertically into metal tubs in a greenhouse. A black polycarbonate sheet with holes cut into it to allow seedlings to emerge is placed over the tub to keep light out. Initially a flatbed scanner was used to take images of the chambers, but for higher throughput of imaging, a box was constructed which holds a root chamber vertically between two digital cameras, which take photos of both sides of the chamber simultaneously [109]. Root angle can then be determined from the photos using software. Using these custom root chambers, four QTLs for nodal root angle in sorghum were identified from a RIL population of 141 lines, derived from BR23296 $\times$ SC170-6-8 [108] (Table 1). The percentage of variance explained by the QTLs (R^2) ranged between 6.72 and 29.78%, and the heritability for the trait was high (73.7%). Heritability for root angle was much greater than for the other traits measured such as root dry weight, shoot dry weight, and total leaf area (13.0, 21.6, and 20.4%, respectively). Some QTL were also identified for these traits, despite the relatively low heritabilities.

Using larger rhizotrons (240 cm wide $\times$ 10 cm thick $\times$ 120 cm deep) with a clear panel on the front, the nodal root angle of sorghum seedlings was shown to indeed correlate to root architecture in mature plants [105]. The narrow-angle genotype BR23296 had more roots visible at 100 cm depth directly below the plant, whereas the wide-angle genotype SC170-6-8 had more roots visible at 80 cm depth at a distance of 120 cm away from the plant. Soil cores were taken from the rhizotrons at harvest, and slight (though not statistically significant) differences were noted in the distribution of water extraction between the narrow and wide-angle genotypes.

An above-ground rhizotron constructed using cement blocks (25 m $\times$ 4 m $\times$ 1.5 m high) was filled with soil [110]. Into this were planted 184 RI lines, and the two parental lines (E36-1 and SPV570), which had contrasting phenotypes for root system size. After 110 days, one wall of the rhizotron was removed, and each plant was carefully removed from the soil with the root system intact. Roots were washed, and then various measurements were taken, including root length, number of roots per plant, root volume, fresh and dry weight, and root to shoot ratio. A genetic map containing 104 markers was constructed, and QTLs were identified on chromosomes SBI-01, SBI-04, and SBI-10 (Table 1) using composite interval mapping.

It is virtually impossible to excavate the entire root system of every plant in a field experiment, particularly when the plants are mature. However, the angle of the crown roots has been studied as a proxy for rooting depth. A population from a cross between drought-tolerant Early Hegari-Sart and drought-sensitive BK7

was developed [69]. The drought-tolerant parent has a steeper crown root angle and exhibits less wilting during dry periods. The mapping population was planted at two locations in Florida, and the roots were dug up at maturity using a technique similar to the “shovelomics” methods developed for maize [104]. After washing away the soil, the crown root angles were measured with a digital protractor. A high-density linkage map was constructed using data from over 6000 GBS-based SNP markers [63]. One QTL was identified for crown root angle on SBI-03 (Table 1). The sorghum genome sequence [56] combined with a very dense genetic map allows for the identification of candidate genes underlying QTLs. The QTL region on SBI-03 for crown root angle contains 85 genes. Among them is a putative tryptophan aminotransferase, which may function in auxin metabolism, and is homologous to a rice gene involved in root gravitropism [69].

The primary function of brace roots is to help stabilize the plant [111]. The relationship between brace roots and below-ground nodal roots has not been studied in sorghum, but in maize, a moderate correlation ($r = 0.53$) between brace root angle and crown root angle was found [104]. The underlying genetics of brace roots have been investigated in sorghum [111]. Sansui, a tall Chinese landrace with numerous brace roots on the lower 6–8 nodes, was crossed with Jiliang 2, an elite variety with brace roots only on the lowest node. The F_1 hybrid had a similar number of nodes with brace roots as Sansui. Two hundred nineteen F_2 plants were analyzed in the field for number of nodes with brace roots, along with other agronomic traits. A genetic map was constructed using 109 polymorphic SSR markers. Using inclusive composite interval mapping, two QTLs were identified on SBI-07 and SBI-06 which explained, respectively, 52.5 and 7.6% of the phenotypic variation in the number of nodes with brace roots (Table 1). These QTLs do not overlap with other root architecture QTLs identified in sorghum, but they were also identified in completely different genetic backgrounds.

Silicon aggregates have been found in the endodermis of sorghum roots, and it has been suggested that this element contributes to the crop’s ability to withstand water stress [73, 112, 113]. Greater silicon content in roots of a drought-tolerant cultivar (Gadambalia, 3.5% Si) was observed as compared to a drought-sensitive cultivar (Tabat, 2.2% Si) [114]; however, it is not clear if their difference in drought tolerance is due to the difference in root silicon content. No other studies on genetic variation for silicon content in sorghum roots have been reported, so it is not certain if this trait could be a target for breeding. Based on homology with maize, three silicon transporter genes (*SbLsi1*, *SbLsi2*, and *SbLsi6*) have been identified in sorghum. Two of these genes (*SbLsi1* and *SbLsi2*) were shown to be highly expressed in the growing root tip

in the presence or absence of silicon. *SbLsi6* showed a lower level of expression, but was slightly upregulated in the absence of silicon [113].

6.2 Drought Tolerance Traits

6.2.1 Compatible Solutes

One means by which a plant cell can cope with osmotic stress brought about by drought or high salinity is to increase the concentration of osmotically active solutes within the cell, thus lowering its osmotic potential [115]. This causes water to flow in, thus maintaining cell turgor and normal functions [116]. Compatible solutes are osmotically active small molecules that can accumulate to high concentrations within the cell without interfering with normal cellular functions [115]. They may also have protective or stabilizing effects on enzymes and membranes [116, 117]. Proline [118–120] and glycinebetaine [122] have been identified in sorghum as compatible solutes that may confer some level of tolerance to drought. Variation in the ability to accumulate glycinebetaine has been identified in both maize and sorghum [121, 122]. In sorghum, IS2319 has been identified as glycinebetaine-deficient, while P932296 is a glycinebetaine accumulator. A RIL population from a cross between these two lines was developed [117]. The concentration of leaf glycinebetaine showed a segregation pattern which suggested one major gene determines whether a line produces glycinebetaine, as well as one or more genes of smaller effect which control the amount of glycinebetaine in lines that do accumulate this solute. In the F_{7:8} generation, 10 individual plants from 28 RILs were screened for differences in glycinebetaine accumulation, to identify those that were still segregating for the trait. From two segregating families, two pairs of NILs were derived for further study. Biochemical analysis suggested that the deficiency in glycinebetaine was caused by the inability to convert choline to betaine aldehyde, the immediate precursor to glycinebetaine. This step is catalyzed by choline monooxygenase (CMO); however, no difference in the expressed CMO sequences or level of gene expression between pairs of accumulating and non-accumulating NILs was found [123]. No QTLs for compatible solute accumulation have yet been reported in sorghum.

6.2.2 Pre-flowering Drought Tolerance

Pre-flowering drought stress in grain sorghum occurs when plants are experiencing significant water stress prior to anthesis, especially from panicle differentiation to anthesis. Plants that are under pre-flowering drought stress will exhibit symptoms of leaf rolling, leaf erectness, reduced height, delayed flowering, panicle blasting, saddle effect, and reduced panicle size [124]. Figure 1 demonstrates pre-flowering drought stress tolerance and susceptibility in Lubbock, TX in June 2017. The evaluation of pre-flowering drought tolerance is also confounded by plant maturity; early maturing genotypes often escape water stress and consistently express a tolerant phenotype. Sources of pre-flowering drought



Fig. 1 Pre-flowering drought tolerance (left) versus pre-flowering drought susceptibility (right). Picture taken in Lubbock, TX in June 2017

tolerance have been identified, and have been used in sorghum breeding programs for many years [124, 125].

Compared to post-flowering drought tolerance, genetic mapping studies identifying QTLs for pre-flowering stress tolerance have been somewhat limited. One of the most important RIL populations used in the genetic study of pre-flowering stress tolerance to date is “Tx7078 $\times$ B35.” Six QTLs were identified that were specifically associated with pre-flowering drought tolerance [125] (Table 1). The alleles derived from Tx7078 were associated with enhanced drought tolerance at all but one QTL. Tolerance measured as grain yield per se was 42.1% of the total phenotypic variation explained, indicating that pre-flowering drought tolerance plays an important role in overall grain yield in water stress environments. Plant phenology also strongly interacts with drought tolerance and grain yield [126]. In the pre-flowering drought tolerance study, QTLs for maturity and height were often co-localized with QTL for drought tolerance [125].

Kebede et al. [127] utilized an F_7 RIL population derived from the cross SC56 $\times$ Tx7000 to identify QTL for both pre-flowering drought tolerance and post-flowering drought tolerance. Three

QTLs for pre-flowering drought were identified (Table 1). *Pfr G* was identified as a major QTL, contributing 15% and 37.7% of the phenotypic variance in two environments [127]. Most of the pre-flowering QTLs were unlinked to post-flowering drought tolerance and other agronomic QTLs; therefore, incorporation and pyramiding of both types of drought tolerance should be possible.

6.2.3 Post-flowering Drought Tolerance (Stay-Green)

Post-flowering drought tolerance, also known as stay-green, is an agronomically important drought tolerance trait that produces greater retention of green leaf area from anthesis to grain-fill maturity [128–130]. Stay-green has been important in sorghum production regions throughout the world that experience consistent water-limited conditions during the grain fill period. For example, Australia and other Mediterranean climates have utilized the stay-green trait extensively because producers routinely rely on early season rains and stored soil moisture to sustain a crop to maturity [131]. The phenotype is associated with greater green leaf area, reduced lodging, increased basal stem sugars, reduced susceptibility to charcoal rot (*Macrophomina phaseolina*), increased nitrogen content, and increased seed size at maturity [129, 132]. Figure 2 demonstrates post-flowering drought tolerance versus susceptibility in a breeding nursery in Corpus Christi, TX in 2015. Prior studies also identified a link between crop nitrogen content, extent of N-mobilization post anthesis, and the onset of leaf senescence associated with differences in stay-green [133–135]. The value of the stay-green trait in terms of overall grain yield is often debated



Fig. 2 Post-flowering drought tolerance (left) versus post-flowering drought susceptibility (right). Picture taken in Corpus Christi, TX in August 2015

and in many cases, is the result of trade-offs during the growing season between increased production in water-limited seasons and decreased overall production in high-yielding seasons. Jordan et al. [131] identified that stay-green on average was positively associated with grain yield in environments that yielded 6 metric tons per hectare or less. In high yielding environments, the positive attributes of the stay-green trait declined and in some environments, were associated with a reduction in grain yield. Nevertheless, the stay-green trait appears to be an important drought tolerance trait that often produces higher, and more stable yields, especially since sorghum is primarily produced on marginal lands where optimal growing conditions are seldom realized.

Many RIL populations have been utilized to map stay-green QTLs in sorghum [127, 136–142]. The four parental genotypes from these studies that confer stay-green are BTx642, QL41, SC56, and E36-1. A list of 39 QTLs ($\text{LOD} > 2$) that were identified from five studies were compiled onto a single genetic map and 12 regions of the sorghum genome that contain stay-green QTLs were identified [143]. Favorable alleles for stay-green from more than one genotype were identified in many of the regions from multiple studies which gives further support of the importance of these QTLs. Yet, to date, no genes have been identified from any of these 12 regions.

One of the most widely utilized stay-green RIL populations for many research groups is BTx642 $\times$ Tx7000, which was originally developed by Texas A&M University in the 1990s [141]. BTx642 (also known as B35) is a widely used stay-green line for both stay-green mapping and breeding. BTx642 was derived from a BC₁ selection of BTx406 and an Ethiopian durra line, IS12555, has excellent stay-green ratings, and has been used extensively for drought tolerance breeding [130, 140]. Studies have shown that the genes conferring stay-green act with varied levels of dominance and partial dominance [144], which is an important trait for the production of stay-green hybrids. Four original stay-green QTLs (*Stg1*, *Stg2*, *Stg3*, *Stg4*) were mapped using the BTx642 $\times$ Tx7000 population [141] and their relative ranking based on contribution to the stay-green phenotype are *Stg2* $>$ *Stg1* $>$ *Stg3* $>$ *Stg4*. *Stg1* and *Stg2* are located on SBI-03, *Stg3* on SBI-02, and *Stg4* on SBI-05 [145, 146] (Table 1).

NILs of the non-stay-green line Tx7000 carrying each individual *Stg* allele have also been developed and results demonstrate that each *Stg* allele (*Stg1*-*Stg4*) reduced leaf senescence in the Tx7000 genetic background. Among the four *Stg* alleles, NILs containing alleles of *Stg2* had the highest green leaf area at maturity (GLAM) among all NILs analyzed, further supporting that *Stg2* has the greatest influence on the stay-green phenotype [147]. Borrell et al. [148] also investigated the mechanisms that control the

stay-green trait in sorghum using the Tx7000 NILs which contain individual *Stg* QTL and concluded that *Stg* loci reduced the overall canopy size at flowering, reduced tillering, and promoted overall root growth. A reduction in total canopy size at anthesis reduced overall water demands of the plant, and allowed for greater water availability during grain filling. A positive correlation between canopy size and water use at anthesis was observed resulting in less water use in three of the four Tx7000 *Stg* NILs compared to Tx7000. Therefore, increased grain yields associated with *Stg* loci are at least partially explained by increased water use post-anthesis in response to an overall reduced canopy size [148].

Recently, a novel stay-green QTL, designated *Stg5*, has been identified on SBI-01 that co-localizes with known dhurrin biosynthetic genes [149]. Dhurrin is a cyanogenic glucoside present in sorghum that once catabolized by the enzyme dhurrinase, produces hydrogen cyanide (HCN) gas as a defense mechanism against herbivory [150]. Hayes et al. [149] identified *Stg5* on the distal arm of SBI-01 using a new digital genotyping (DG) genetic map of BTx642 × Tx7000 with a total of 897 unique markers spanning all ten chromosomes (Table 1). In addition, *Stg1-Stg4* were also re-identified in the study, confirming that stay-green is a complex trait with many potential mechanisms. Although it still remains unclear how exactly dhurrin is associated with the stay-green phenotype, it is hypothesized that dhurrin could be providing an N source for grain filling under water limiting environments, therefore delaying plant senescence. The increased production of the N rich compound dhurrin could also be altering leaf growth and development by affecting N-status or signaling in developing leaves, thus suggesting that elevated levels of leaf dhurrin could cause a reduction in leaf area, a key trait associated with the function of *Stg1-Stg4*.

Furthermore, in a greenhouse study, drought stress was imposed beginning at the boot stage in a diverse set of 107 sorghum accessions [151]. After 12 days, the percentage of dry leaves was recorded in the drought-stressed plants. A QTL for percentage of dry leaves was identified by association analysis on SBI-01 (Table 1), which may correspond to the previously identified stay-green QTL *StgF* [136], or *StgA* [142]. However, only 98 SSR markers were used in this study, so the power to detect QTLs in this study may have been limited.

7 Future Perspectives

The use of aquifers to irrigate crops began in the second half of the twentieth century, but in Asia, Africa, Australia, and North America the withdrawal of water has exceeded the recharge causing a depletion of water in the aquifers [45]. Four areas of the U.S experience

groundwater depletion of which the Ogallala Aquifer, which exists under the central portion of the U.S., is depleted in Kansas and Texas [45]. As water becomes a limiting factor due to depletion of aquifers, a growing world population, water pollution, and climate change, more drought-resistant crops must be grown that can exist without irrigation. Sorghum is well adapted to hot and dry environments, is frequently grown without irrigation [3], and has a large number of genetic resources for continual drought tolerance improvement [152]. Utilization of genes in sorghum hybrids that confer pre-flowering and post flowering drought tolerance, enhance root systems, promote thick cuticles and trichome development, and enhance water use efficiency may help meet the need of growing crops in warmer and water restricted environments. Yet the effect of these genes must be studied for their impact on sorghum yield and/or growth under normal and drought stress conditions as selection for drought resistance traits can come at a cost to yield. Using this information, sorghum breeders must combine the most effective QTLs or genes controlling drought response traits into hybrids without a negative effect on yield under non-drought conditions.

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